

COSC2148/COSC3130 Computing Research and Project Preparation

Assignment 2

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Section 1

Introduction

Widespread misinformation, generating challenges for information credibility alongside informed decision-making can be attributed to the rise of social media. The increasing complexity of online content consisting of textual, visual, and interaction-based elements, has increased the difficulty of accurately identifying false information. Thus, the development of reliable misinformation detection systems has become a critical research focus.

Early research in misinformation detection has been predominantly characterised using text-based classification models, where linguistic features are utilised to determine content authenticity. However, limitations in these approaches have been observed, predominantly in the inability to capture non-textual signals embedded within modern social media content. Multimodal learning has partially overcome this limitation by integrating text alongside images. For example, EANN framework demonstrated that combining both text and images leads to enhanced detection performance compared to unimodal approaches (Wang et al., 2018). Similar to this, large-scale multimodal dataset such as Fakeddit allows better testing, showing that multimodal representations provide superior accuracy when detecting complex misinformation classes (Nakamura et al., 2020). Moreover, recent progress in multimodal architectures has been made towards cross-modal interaction and multimodal feature fusion. Models such as HMCAN and MPFN prove that modelling relationships between text and image modalities significantly enhances detection performance (Qian et al., 2021; Jing et al., 2023).

In parallel, the incorporation of social context information has been explored through the development of context-aware detection models. These approaches utilise features derived from user interactions, propagation patterns, and temporal dynamics to improve classification performance. For example, models using graph representations of temporally evolving graphs to represent social media dynamics allow for the modelling of information diffusion in the social media environment (Song et al., 2021). On the other hand, models based on social interactions between posts and social media users have shown the ability of social interactions to be employed to detect misleading posts (Min et al., 2022). Moreover, recent advances have demonstrated that incorporating both modalities of content and social context data enhances the performance of the classifier (Nadeem et al., 2023; Jing et al., 2025).

Despite these advancements, a crucial limitation remains being the lack of systematic integration between multimodal content modelling and context-aware architectures. Existing

context-aware frameworks (i.e. DANES and GETAE), retain a text-only content branch restricting their ability to fully leverage multimodal signals present in contemporary social media environments (Truică et al., 2024). Whilst prior studies have explored the combination of multimodal content and social context, limited research has conducted controlled evaluations that isolate the contribution of each component within a unified framework (Dellys et al., 2025). This results in an incomplete understanding of the relative impact of multimodal content features and social context features on detection performance.

This research proposes the development of a context-aware multimodal misinformation detection framework, in which the traditional text-based content branch is substituted with a multimodal content branch that integrates both textual and visual representations. The research adopts a comparative experimental design to enable the systematic evaluation of architectural modifications through controlled substitution and ablation-based analysis. The objective of this research is to determine whether the integration of multimodal features within a context-aware framework results in measurable performance improvements, and to analyse the relative contribution of multimodal content and social context components to overall detection effectiveness.

The contribution of this research is threefold. First, existing context-aware architectures are extended through the incorporation of a multimodal content branch. Second, a structured experimental framework is introduced to enable direct attribution of performance improvements through controlled ablation. Third, empirical insights are generated into the interaction between multimodal content features and social context signals. The significance of this research lies in its potential to support the development of more robust and contextually aligned misinformation detection systems that reflect the complexity of real-world social media environments.

The structure of this paper is organised as follows. In Section 2, the research questions are presented, which provide a basis for the problems to be solved through this study. Section 3 is concerned with an analysis of previous research, particularly on multimodal and context-aware systems of misinformation detection, highlighting their limitations. The research methodology, comprising details on system architecture, experiment setup, and evaluation method, is described in Section 4. Section 5 deals with describing the procedure for evaluating the performance of our model against previous benchmark models. Finally, Section 6 explains the approach adopted for project management.

1.1 Research Questions

This research aims to develop a context-aware misinformation detection framework that substitutes the text-only content branch used in established context-aware ensemble models, such as DANES [1] and GETAE [2], with a multimodal content branch that jointly captures textual and visual information. While recent work has shown that multimodal content features and social context features each independently improve fake news detection, the closest existing combinations [3, 4] use social branches that differ architecturally from the propagation-graph ensembles of the DANES/GETAE family. As a result, two questions remain unanswered, and together they form the technical core of this project.

RQ1. *How does replacing a text-only content branch with a multimodal content branch affect the performance of context-aware inauthentic social media content detection models, and to what extent?*

What it addresses. This question isolates a single architectural change: keeping the social context branch of DANES and GETAE intact while substituting the text-only content processor with a content branch that fuses textual and visual features (e.g., BERT for text and a CNN- or transformer-based visual encoder for images). The effect is measured against the original text-only baselines on the same benchmark datasets used in the DANES and GETAE papers (Twitter15 and Twitter16), as well as the multimodal Fakeddit dataset [5], using accuracy, F1-score, precision, and recall.

Significance. Misinformation on social platforms increasingly relies on manipulated images, memes, and screenshots, where the textual signal alone is insufficient or misleading [6]. Even strong context-aware ensembles inherit the blindness of their text-only content component, meaning a measurable portion of detectable misinformation may be missed not because the social signal is weak, but because the content signal is incomplete. Quantifying the gain (or absence of gain) from a multimodal substitution directly informs whether the established DANES/GETAE family of architectures is leaving accuracy on the table.

Novelty and complexity. No prior work directly performs this controlled substitution and isolates its effect on the DANES/GETAE backbone. Existing multimodal-plus-context models such as EFND [7], DPSG [8], and the framework of Dellys et al. [9] use fundamentally different social branches (engagement signals, dynamic propagation graphs, or attention-based fusion), so their results cannot be cleanly attributed to the multimodal component. Answering RQ1 requires re-implementing a non-trivial baseline, designing a compatible multimodal branch, and ensuring the comparison remains controlled — a challenging but well-defined contribution.

1.2 Research Questions

Research questions - 1 page (4 marks)

This section should address the technical issues you will be addressing in your research. It is about the “what” (you will be addressing/focussing when you start your thesis).

Describe here the specific research questions to be addressed. You need to break the problem

you are addressing into specific sub-problems (called here research questions) that could help you during the thesis to come up with innovative solutions properly.

Show how the research questions are interesting and worth addressing.

Please include at least two (2) research questions, but you are limited to a maximum of three (3) research questions.

For each research question, describe what it is, its significance, and why it is important to address it.

1.3 aaaSection2

1.3.1 aaaSection2Subsection1

aaa

1.4 Research Questions

1.4.1 Research Questions

Overview

- This research focuses on evaluating the integration of multimodal content within context-aware fake news detection models.
- The study aims to investigate how combining multimodal features with social context affects detection performance.
- The research questions are designed to break the problem into key sub-problems related to model performance and component contribution.

RQ1: Effect of Multimodal Content Integration

- How does replacing a text-only content branch with a multimodal content branch affect the performance of context-aware inauthentic social media content detection models, and to what extent?
- Significance:
 - Evaluates whether multimodal representations improve detection accuracy.
 - Provides direct comparison with existing context-aware models such as DANES and GETAE.
 - Helps quantify the performance gain from incorporating visual features alongside text.
- Why it is important:
 - Addresses a key limitation in current models that rely on text-only content.
 - Helps determine if multimodal integration is worth the added complexity.

RQ2: Contribution of Model Components

- How does the relative contribution of the multimodal content branch and the social content branch affect overall detection performance?
- Significance:
 - Analyses the importance of different components within the model.
 - Identifies whether content features or social context features contribute more to performance.
- Why it is important:
 - Supports better design of hybrid detection frameworks.
 - Helps prioritise which components should be optimised in future models.

Section 2

Literature Review

2.1 Overview of Social Media Misinformation Detection

The development of misinformation detection in social media can be observed across three main directions: text-based content analysis, social-context modelling, and multimodal detection. Earlier studies framed misinformation detection as a content classification problem, focusing on the linguistic and semantic properties of posts. While these approaches established useful baselines, increasing limitations have been identified in modern social media environments, where misleading information is conveyed not only through text but also through images, memes, screenshots, and user interaction patterns. As a result, a shift toward multimodal learning and context-aware modelling has been observed, reflecting the need to capture broader signals available within online platforms [Qian et al., 2021; ?; ?].

2.2 Text-Only Content-Based Detection Approaches

Text-only content modelling has remained a foundational component in misinformation detection, including within later context-aware systems. In the relevant literature, post content continues to be represented primarily through text-based features. The main strength of this approach is the provision of a strong semantic baseline, particularly when misinformation is expressed through written language. However, limitations are increasingly evident in social media contexts, where misinformation is often conveyed through visual and multimodal formats. This is visible in models such as DANES and GETAE, where text-only representations are still used within the content branch despite strong performance [??]. Thus, while useful as a baseline, text-only modelling provides a restricted view of content and is insufficient for capturing the full complexity of contemporary inauthentic posts.

2.3 Context-Aware Detection with Social and Propagation Signals

A major research direction involves the incorporation of social context into misinformation detection. Rather than focusing only on post content, these approaches model information spread, user engagement, and temporal interaction patterns. Evidence shows that diffusion

patterns, user relationships, and temporal dynamics provide strong classification signals, often capturing behavioural evidence unavailable from content alone [???]. Models such as DANES and GETAE represent important context-aware ensemble architectures, combining a content branch with a social or propagation branch. Their strength lies in structured use of social context and strong benchmark performance. However, reliance on a text-only content branch limits the ability to capture visual misinformation. Additionally, propagation-based methods depend on interaction data that may be sparse or delayed, particularly in early detection scenarios. Consequently, while detection extends beyond content, limitations of text-only representation are still inherited [??].

2.4 Multimodal Content-Based Detection Approaches

Multimodal detection addresses the limitations of text-only modelling. Foundational work such as EANN demonstrated improved performance through combining textual and visual features [?]. Later studies focused on improving fusion methods. For example, MCNN models text-image consistency, while HMCAN and MPFN show that hierarchical attention and progressive fusion better capture cross-modal relationships than simple concatenation [?Qian et al., 2021; Jing et al., 2023]. This progression indicates that gains depend not only on modality inclusion but also on interaction quality. Benchmark datasets such as Fakeddit further show that multimodal classifiers outperform unimodal baselines [??]. While this literature reflects modern social media content well, a key limitation is the focus on content-level modelling, with limited integration of social and propagation context.

2.5 Integrating Multimodal Content and Social Context

Recent research has explored integrating multimodal content and social context within unified frameworks. Studies such as EFND, C²DCMTN, DPSG, TMEF-BI, and ESDGFN show that combining textual, visual, and contextual signals can achieve strong performance [???????]. These findings indicate that multimodal content and social context are complementary. However, variation in how social context is represented introduces challenges in comparison. Some models use engagement data, while others incorporate behaviour, stance, or dynamic graph fusion [????]. This lack of consistency prevents clear attribution of performance gains, as both multimodal and social components are often changed simultaneously. As a result, while joint modelling is supported, the specific contribution of multimodality remains unclear, particularly compared to DANES and GETAE architectures [???].

2.6 Research Gap and Positioning of the Proposed Study

A central gap in the literature is identified. Prior research shows that both multimodal content and social context improve misinformation detection [?Qian et al., 2021; ?; ?; ?]. However, limited research performs controlled substitution of the text-only content branch within established context-aware ensembles while keeping the social branch unchanged. As a result, the specific contribution of multimodal substitution within DANES/GETAE remains unclear, and the relative contribution of content and social context is not well decomposed [????]. This gap aligns with the project’s research questions. Research Question 1 evaluates

performance improvement from multimodal substitution, while Research Question 2 examines whether multimodal and social branches provide complementary or overlapping value. To address this, the study uses DANES and GETAE as baselines, replaces the text-only content branch with a multimodal alternative, and conducts ablation-based comparisons across configurations. This enables a controlled evaluation of the contribution of multimodal content and social context in misinformation detection.

Table 2.1: Summary of prior studies on multimodal and context-aware misinformation detection

Study	Focus	Key finding / contribution	Main limitation
Wang et al. (2018) – EANN	Early multimodal fake news detection	Showed that combining text + image features improves misinformation detection over text-only methods	Fusion was still relatively early-stage and did not fully model deeper cross-modal relationships
Nakamura et al. (2020) – Fakeddit	Multimodal benchmark dataset	Provided a large benchmark showing that multimodal classifiers outperform unimodal baselines	Focused more on benchmarking than integrating social context
Xue et al. (2021) – MCNN	Text-image consistency modelling	Demonstrated that modelling the relationship between text and image improves classification	Mainly content-focused, with limited use of propagation or user context
Qian et al. (2021) – HMCAN	Hierarchical multimodal attention	Showed that attention-based multimodal fusion captures cross-modal interactions better than simple feature concatenation	Still concentrated on content-level signals rather than broader platform context
Song et al. (2021)	Social/propagation context	Found that diffusion patterns and user interaction signals are useful for misinformation detection	Reliant on interaction data, which may be sparse in early detection
Min et al. (2022)	Temporal and behavioural signals	Confirmed that temporal dynamics and engagement behaviour can improve classification	Less effective when user activity data is limited or delayed
Segura-Bedmar & Alonso-Bartolome (2022)	Multimodal benchmarking	Reinforced evidence that multimodal approaches outperform text-only models on social media misinformation tasks	Does not resolve how multimodality compares when social context is held constant
Nadeem et al. (2023) – EFND	Multimodal + contextual integration	Showed that combining content and contextual signals can produce strong detection performance	Hard to isolate whether gains come from multimodality or the added context

Study	Focus	Key finding / contribution	Main limitation
Jing et al. (2023) – MPFN	Progressive multi-modal fusion	Demonstrated that better fusion design, not just adding more modalities, improves results	Limited social-context integration
Truică et al. (2024) – DANES	Context-aware ensemble model	Combined a text-based content branch with a social/propagation branch, achieving strong benchmark performance	Content branch remains text-only, so visual misinformation may be missed
Wang et al. (2024)	Joint multimodal-context modelling	Supported the view that multimodal content and social context are complementary	Social context representation differs from other studies, making comparison difficult
Zhu et al. (2024)	Propagation-aware detection	Showed that behavioural and diffusion evidence adds strong classification value beyond content alone	Performance depends on availability of network/interaction data
Truică et al. (2025) – GETAE	Context-aware ensemble model	Similar to DANES, achieved strong results by combining text content + social context	Still inherits the weakness of a text-only content branch
Jing et al. (2025)	Unified multimodal-context model	Reported strong performance from joint modelling of text, visuals, and context	Multiple components change at once, so the exact source of improvement is unclear
Chen et al. (2025)	Behaviour/context-enhanced multi-modal model	Found that integrating behavioural context with multimodal content improves detection	Comparison across studies is difficult because context features vary widely
Sun et al. (2025)	Dynamic graph/context fusion	Showed value in modelling dynamic relationships and graph-based context alongside content	Adds complexity and reduces comparability with simpler baselines
Dellys et al. (2025)	Comparative/context discussion	Highlighted the difficulty of identifying the specific contribution of multimodal substitution in context-aware systems	Does not fully isolate multimodal effects within DANES/GETAE-style architectures
Ji et al. (2026)	Recent integrated framework	Further supported the effectiveness of combining multimodal and contextual signals	The separate effect of multimodality is still not clearly decomposed

Section 3

Research Methodologies

3.1 Research Design

Building on the existing DANES and GETAE ensemble architectures for inauthentic content detection, the proposed MACE (**M**ultimodal **A**nd **C**ontext-aware **E**nsemble) framework adopts a comparative experimental design to evaluate the impact of integrating multimodal content features into existing ensemble models.

3.2 System Architecture

DANES and GETAE are ensemble approaches fusing a text-based content branch and a social context branch to classify inauthentic content on social media.

HMCAN(Hierarchical Multi-modal Contextual Attention Network Qian et al. [2021]) and MPFN (Multimodal Progressive Fusion Network Jing et al. [2023]) are multimodal architectures performing inauthentic content detection based on textual and visual features in social media posts.

The proposed MACE framework retains the dual-branch structure and social context features of DANES and GETAE and replaces the text-based content branch with a multimodal branch (HMCAN or MPFN¹) content branch, allowing for evaluation of the classification performance of multimodal content features within existing ensemble frameworks.

Diagram Here (TODO: insert architecture diagram showing the proposed modifications to the baseline architecture).

3.2.1 Multimodal Content Branch

MACE adopts HMCAN as the multimodal content branch, which is selected for its task alignment, dataset compatibility and ease of integration with the existing DANES/GETAE architectures.

¹If HMCAN proves incompatible with the selected dataset or integration constraints, MPFN is identified as a potential fallback candidate, utilising a more complex image pipeline.

HMCAN has demonstrated excellent accuracy (89.7%, outperforming common baselines such as EANN) in multimodal fake news detection using its BERT + ResNet encoding pipeline, is consistent with the text branch of DANES/GETAE and has been independently validated on the Multimodal Twitter dataset, making it a suitable candidate.

3.2.2 Social Context Branch

We retain the Graph-based Deep Learning social context branches from the baseline DANES/GETAE architectures.

FOR DISCUSSION: Should we create a RQ to find out if these ensemble architectures can be adapted to a different dataset? we might find that this is not possible without significant modification, and this would have implications for the rest of the study. (This would be an interesting finding in itself), perhaps evaluating different datasets with these models might be an alternate research path? TK

	DANES	GETAE
Social context approach	Graph embedding from static network structure	GNN over dynamic propagation graph
Key information used	User interactions, relationships, and post behaviour	Information diffusion and propagation patterns

Table 3.1: Comparison of DANES and GETAE social context branches.

3.2.3 Feature Fusion and Classification

Both fusion models combine branch outputs by concatenating the content and context embeddings to produce a classification embedding, which is passed to a classification layer. The classification layer applies a softmax function to produce a probability distribution over the classes (authentic or inauthentic) and assigns a label based on the highest probability.

MACE retains this fusion mechanism to ensure the performance differences can be attributed to the substitution of the content branch rather than changes in the fusion or classification approach.

3.3 Experimental Design

The experiment design is structured as a two-stage process to systematically evaluate the proposed architecture and address the research questions. Both base architectures are evaluated allowing the findings to be evaluated for consistency across different fusion approaches.

The model is treated as a replication factor, rather than a variable, as the primary focus is on evaluating the impact of the proposed architecture modifications (multimodal content branch), rather than comparing the performance of the two fusion approaches.

3.3.1 Experiment Stages and Conditions

Stage 1 is to replicate the published architectures and evaluate their performance on the Twitter15 and Twitter16 datasets to ensure the implementation is correct and consistent with the published results.

Stage 2 evaluates four conditions on a multimodal dataset. Condition A establishes a new baseline and Conditions B, C and D systematically evaluate the contribution of content and context features against this new baseline to isolate the impact of the proposed architecture modification.

Stage	Label	Content Branch	Context Branch	Goal / Purpose
1	A	Implementation Baseline	Text-only ✓	Replicate baseline fusion models and confirm published results prior to introducing modifications.
2	A	Dataset Baseline	Text-only ✓	Establish baseline on multimodal dataset using existing fusion models.
	B	Ablation 1	Text-only ×	Isolate contribution of text-only features.
	C	Ablation 2	Multimodal ×	Isolate contribution of multimodal features.
	D	Proposed (MACE)	Multimodal ✓	Evaluation of full proposed framework.

Table 3.2: Summary of experiment conditions and objectives.

3.3.2 Research Question Alignment

RQ1 is directly addressed by comparing the performance of the proposed architecture (Condition D) against the baseline architecture (Condition A) with the multimodal dataset.

A pairwise comparison of condition’s B, C and D address RQ2 by evaluating the contribution of the multimodal content branch and the social context branch to the overall performance of the model.

	Comparison	What varies	What it tells us
RQ1	A vs D	Content branch	Does substituting the multimodal branch for the text branch improve performance within the full context-aware framework?
RQ2	A vs B	Context branch	How much does the social context branch contribute with text-only content?
	B vs C	Content branch	Does adding visual features improve detection independent of social context?
	C vs D	Context branch	How much does the social context branch contribute with multimodal content?

Table 3.3: Pairwise condition comparisons and research question alignment.

3.4 Project Timeline

Section 4

Evaluation

4.1 DELETE original question

evaluation - 1 page (3 marks)

This section is important and should clearly show the evaluation plan of your research. This evaluation plan could be used later during your thesis to clearly demonstrate that you made research contributions in the field of study.

This section will need to address the following points:

- Benchmarking: clearly select some existing methods/algorithms you will need to compare your research work with. It is important to clearly explain these methods/algorithms as well as justify the reasons for choosing them.
- Data sets: describe at least two well-accepted data sets that you will be using to compare your research work against the selected benchmarked methods/algorithms. Again, you will need to justify the reasons for selecting such data sets.
- Metrics: to compare your research work with benchmarked methods/algorithms, you clearly need to list the metrics to be used for this purpose. This could be, for example, "execution time" if you are comparing the performance of your work with other methods/algorithms. You also need to explain why you have selected such metrics. List at least two metrics.

4.2 aaaSection2

4.2.1 aaaSection2Subsection1

aaa

TIMS CONTENT - I ACCIDENTALLY WROTE THIS IN THE METHODOLOGY CHAPTER BUT THEN I REALISED I DIDN'T NEED IT, SO INCORPORATE IF NEEDED I GUESS?

4.3 Datasets and Data Preparation

Twitter15 and Twitter16 are used for Stage 1 replication only, as they do not contain image data.

For Stage 2, a multimodal dataset will be used containing multimodal content (text and images) and social context features through a conversation thread structure, comparable to Twitter15/16 (to ensure the social context branch can be retained with minimal modification).

We are not intending to derive conclusions about the relative performance of the models on the Twitter15/16 datasets and the multimodal dataset so comparisons between stage 1 and 2 are not relevant¹.

TODO this needs fleshing out a bit more I think

4.4 Evaluation Strategy

The four-condition design allows for systematic evaluation of the proposed architecture modification by isolating the impact of multimodal content and social context features through controlled ablation experiments.

Performance is evaluated using accuracy, precision, recall, and F1-score, which are standard metrics for classification tasks and allow for comprehensive assessment of model performance across both classes (authentic and inauthentic).

ROC AUC will also be used to evaluate the model’s ability to distinguish between classes across different thresholds.

¹It is anticipated that the transition from Twitter15/16 datasets to a multimodal datasets will result in a performance difference in Stage 2 Condition A relative to Stage 1 depending on the extent to which these models generalise to a new dataset.

Section 5

Project Management

5.1 DELETE original question

project management - 1 page (1 mark)

This section should outline how you plan to manage your research project throughout the semester.

It is essentially a Project Plan that clearly defines the scope and deliverables, including explicit exclusions.

It should present a structured schedule with milestones and a transparent critical path, while assigning responsibility to specific individuals to ensure accountability.

It must also incorporate risk management to anticipate threats and formulate appropriate responses, protecting project objectives from avoidable failure.

This section should also clearly state the methods you will use to manage the project, such as Agile sprints, Scrum ceremonies, Kanban, etc., and specify the enabling tools (e.g., MS Project, Trello, JIRA, Slack, etc.) that will operationalise these methods.

Briefly explain how these methods and the supporting tools together will facilitate scheduling, progress tracking, communication, and reporting. Specifically, you must include the following:

- High-level scope and deliverables of your research project. Clearly state exactly what you will deliver and explicitly rule out what you will not.
- Timeline with major milestones derived from the Work Breakdown Structure (WBS). This should reflect the schedule of your research project and include any critical paths. Note: you don't need to include the WBS itself.
- Clear assignment (ownership) of each task to project team member(s), not roles, for accountability and monitoring.
- Approach in risk management, identify risks and proposed mitigation and contingency plan, as well as escalation paths. List the major risks, rank them, and state what you will do before, when, and after they occur.

5.2 aaaSection2

5.2.1 aaaSection2Subsection1

aaa

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Appendix A

Generative AI Usage

Declaration

[REPLACE THIS LINE WITH YOUR GENERATED DECLARATION FROM
<https://aiattribution.github.io/>]

Tool Report: [Tool Name, e.g. ChatGPT / RMIT Val / Copilot]

Field	Details
1. GenAI tool used	[e.g. ChatGPT 4o, RMIT Val, GitHub Copilot]
2. Purpose for this assignment	[e.g. Brainstorming research questions, drafting outline, rewriting for clarity, code generation]
3. Prompts or instructions given	[Include key prompts used. If conversation was lengthy, include all your prompts in full and use [...] to truncate AI responses only.]
4. Number of prompts given	[e.g. 7]
5. Summary of GenAI output	[Describe what the AI produced. Include brief excerpts where relevant.]
6. Verification of GenAI output	[Explain how you checked the AI’s output for factual accuracy, source validity, and logical soundness. e.g. “All cited sources were manually located and verified via Google Scholar.”]
7. Usage or modification of output	[Explain how you used, adapted, or discarded the AI-generated content in your final submission.]
8. Reflection	[What worked well? What did not? How did it support your learning? What value did it add?]

Table A.1: Generative AI usage report — [Tool Name]